Ideation Phase
Brainstorm & Idea Prioritization Template

Date	30 September 2026
Team ID	SWTID-2026-7065
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	4 Marks

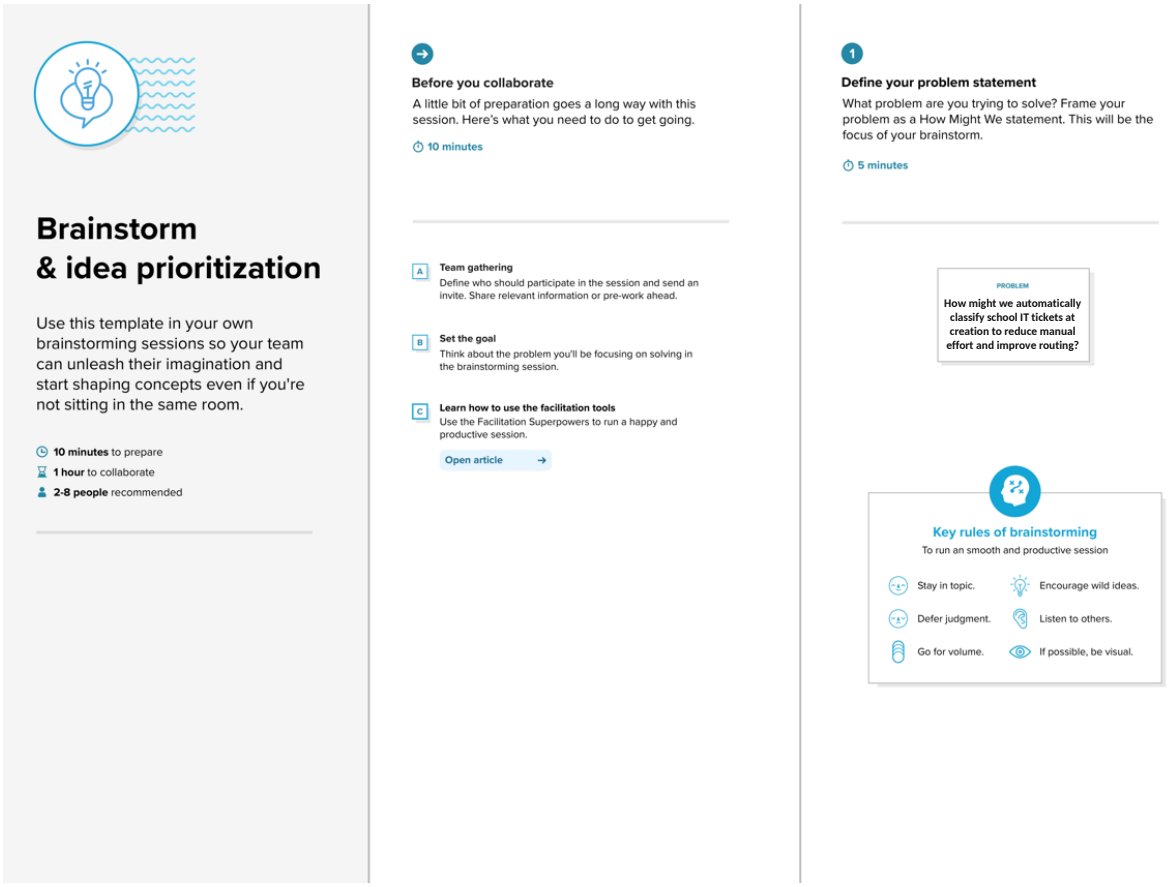
Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

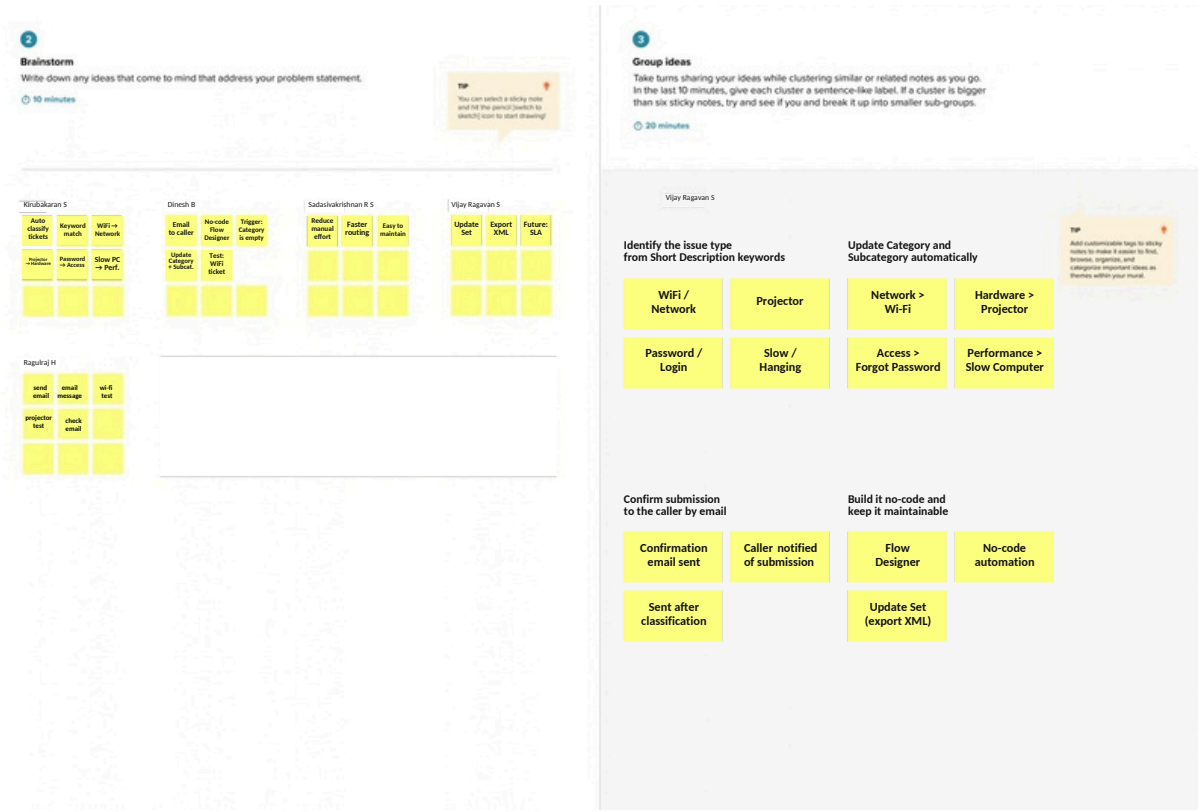
Use this template in your own brainstorming sessions so your team can unleash their imagination and start shaping concepts even if you're not sitting in the same room.

Reference: https://www.mural.co/templates/brainstorm-and-idea-prioritization

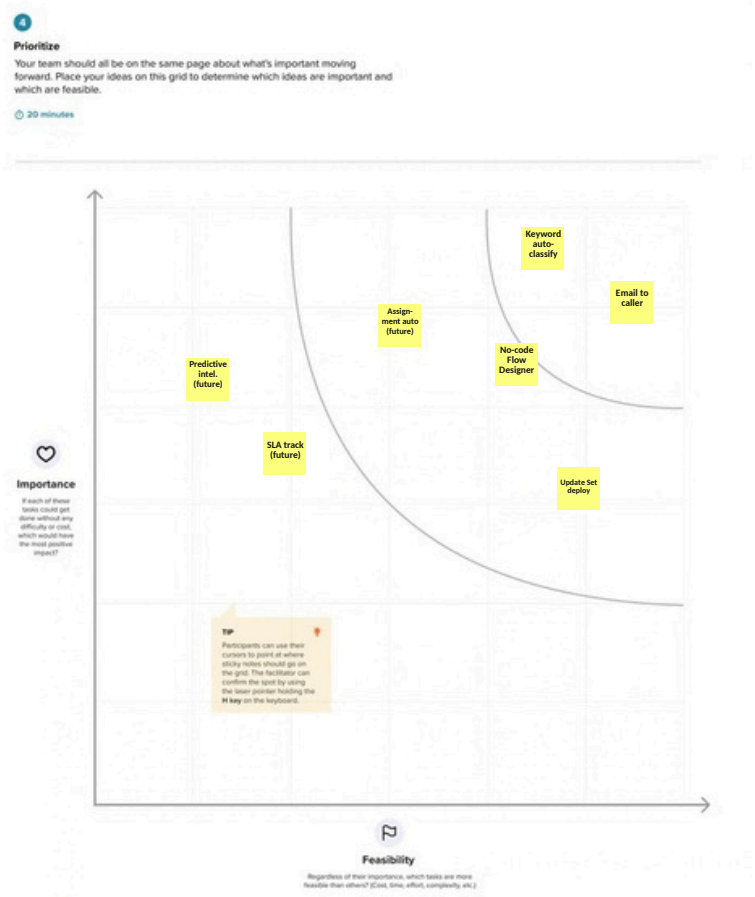
Step-1: Team Gathering, Collaboration and Select the Problem Statement



Step-2: Brainstorm, Idea Listing and Grouping



Step-3: Idea Prioritization



Ideation Phase
Define the Problem Statements

Date	30 September 2026
Team ID	SWTID-2026-7065
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	2 Marks

Customer Problem Statement Template:

Create a problem statement to understand your customer's point of view. The Customer Problem Statement template helps you focus on what matters to create experiences people will love.

A well-articulated customer problem statement allows you and your team to find the ideal solution for the challenges your customers face. Throughout the process, you'll also be able to empathize with your customers, which helps you better understand how they perceive your product or service.

I am	Describe customer with 3-4 key characteristics - *who are they?*	A student or teacher at the school who raises IT support tickets (Wi-Fi, projector, password and slow-computer issues).
I'm trying to	*List their outcome or "job" the care about - what are they trying to achieve?*	Get my IT issue reported and routed quickly to the right support team.
but	Describe what problems or barriers stand in the way – *what bothers them most?*	IT staff review each request and assign the Category and Subcategory manually.
because	Enter the "root cause" of why the problem or barrier exists – *what needs to be solved?*	There is no automatic classification when a ticket is created, so manual review is time-consuming and inefficient.
which makes me feel	Describe the emotions from the customer's point of view – *how does it impact them emotionally?*	Unsure whether my request was received and frustrated by delays.

Reference: https://miro.com/templates/customer-problem-statement/

Example:

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A student or teacher who raises IT support tickets	Get my IT issue (Wi-Fi, projector, password, slow computer) reported and routed quickly	IT staff review each request and assign the category manually	Tickets are not classified automatically when they are created	Unsure whether the request was received and frustrated by delays
PS-2	An IT helpdesk agent who handles incoming incident requests	Classify and route tickets efficiently with minimal manual effort	Reviewing every ticket and setting Category and Subcategory by hand is time-consuming	There is no automation to identify the issue type from the Short Description	Overloaded by repetitive manual work

Ideation Phase
Empathize & Discover

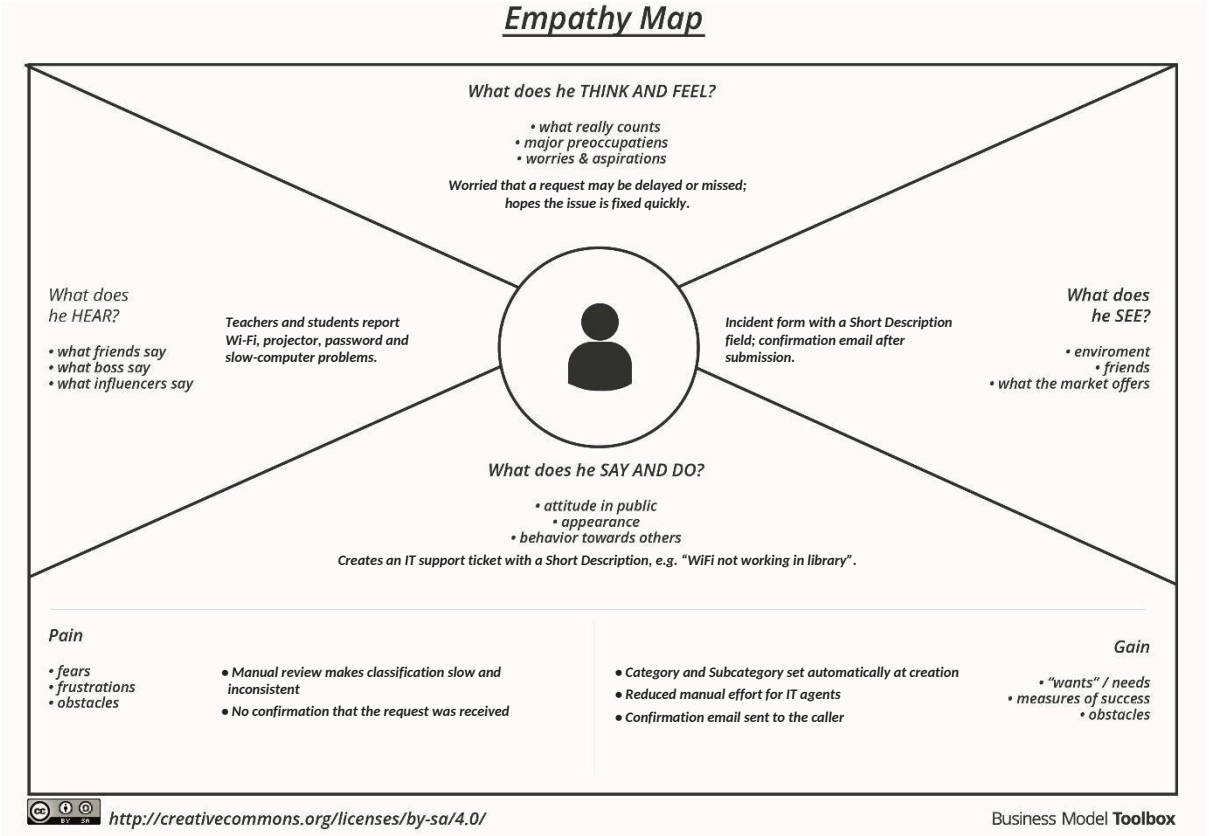
Date	30 September 2026
Team ID	SWTID-2026-7065
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	4 Marks

Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviours and attitudes.

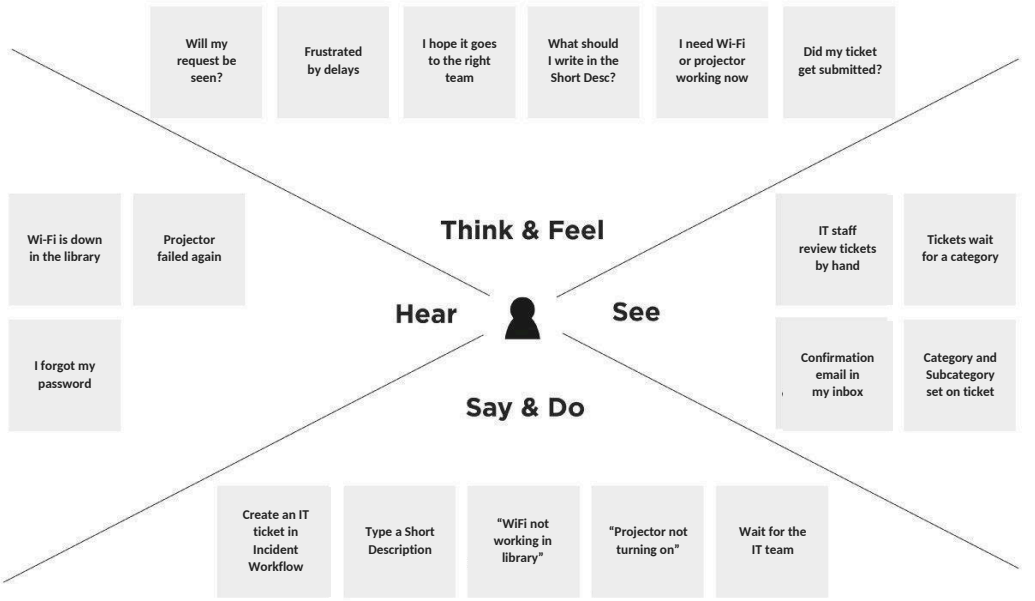
It is a useful tool to helps teams better understand their users.
Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with his or her goals and challenges.

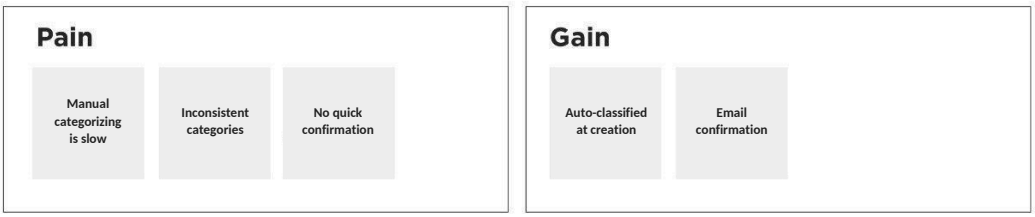
Example:



Reference: https://www.mural.co/templates/empathy-map-canvas

Example: Auto Ticket Classification using Flow Designer





IT HELPDESK

Auto Ticket Classification using Flow Designer

Based on the Auto Ticket Classification using Flow Designer project documentation

Student	Teacher	IT helpdesk staff	IT administrator	Ticket caller

SCENARIO

Reporting an IT issue and getting the ticket auto-classified

Entice

A student or teacher has an IT problem and decides to contact the school IT helpdesk.

Enter

The user opens the IT support ticket form and enters the caller name and short description.

Engage

The ticket is submitted, Flow Designer checks the short description and assigns Category and Subcategory.

Exit

An automated email confirms the request and the ticket is stored with structured details.

Extend

The solution can be extended with assignment automation, SLA tracking and predictive intelligence.

Steps

What does the person (or group) typically experience?

Experience IT problem	Decide to get help	Go to ServiceNow	Find IT support form	Ready to report	Open IT support form	Enter caller name	Enter issue description	Add more details	Skip category fields	Submit ticket	Automatic classification	Category and subcategory assigned	Email confirmation	Email confirmation received	Ticket processed by IT team	Assignment automation	SLA tracking	Predictive intelligence	More classifications
A student or teacher faces a Wi-Fi, projector, password or slow computer issue	The user decides to contact the school IT helpdesk	The user reaches the ServiceNow portal from a computer or mobile device	The user finds the IT support ticket form to report the issue	Only the caller and a short description of the issue are needed	The user opens a new record on the IT support ticket form	The user enters the caller name on the ticket form	The user types a short description, e.g. WiFi not working in library	The user can add extra detail in the Description field	Category and Subcategory are not selected by the user	The ticket is submitted and Flow Designer is triggered	Flow Designer checks the short description for keywords	Category and Subcategory are set, e.g. Network → Wi-Fi	An automated email is sent to the caller confirming the request was submitted	The caller receives the confirmation in their email	The ticket is stored with caller, category, subcategory, state and assignment details	Later, tickets could be assigned to IT groups automatically	The solution can be extended with SLA tracking	Predictive intelligence can be added in future	More keywords and ticket categories can be added to the flow

Interactions

What interactions do they have at each step along the way?

- **People:** Who do they see or talk to?
- **Places:** Where are they?
- **Things:** What digital touchpoints or physical objects would they use?

Computer or mobile device in the classroom, lab or library	IT helpdesk, the team that receives the request	ServiceNow portal on a computer or mobile device	ServiceNow IT support ticket form	Incident Workflow ticket form (New record) Student or teacher in the classroom, computer lab or library	ServiceNow IT support ticket form	Caller field on the ticket form	Short Description field on the ticket form	Description field on the ticket form	Category and Subcategory fields left empty	Submit button on the ticket form Student or teacher submitting from the school campus	Flow Designer (Auto Classify School IT Tickets)	Short Description field checked by Flow Designer Category and Subcategory fields filled by the Update Record action Incident Workflow table updated with the classification	Automated email notification sent by Flow Designer Caller's email address used as the recipient Subject: Your Request for the issue has been submitted.	Caller's email on a computer or mobile device	Incident Workflow table record IT helpdesk staff and IT administrator review tickets	Flow Designer with new keyword conditions IT administrator maintains the flow and keywords	ServiceNow ticket records	ServiceNow with predictive intelligence	Category and Subcategory choice lists

Goals & motivations

At each step, what is a person's primary goal or motivation? ("Help me..." or "Help me avoid...")

Help me report my IT problem easily.	Help me reach the school IT helpdesk.	Help me find the right IT support form.	Help me avoid choosing the wrong category.	Help me report Wi-Fi, projector, password or slow computer issues.	Help me enter my name as the caller.	Help me describe my issue in a few words.	Help me raise a ticket without extra fields.	Help me add more detail if I need to.	Help me submit my ticket with minimal effort.	Help me get my ticket classified automatically.	Help me get the correct Category and Subcategory.	Help IT staff get consistent ticket information.	Help me know that my request was successfully submitted.		Help IT staff process my request faster.	Help IT staff handle more requests as they grow.	Help IT staff track tickets with SLA tracking.		Help the helpdesk add more ticket types later.

Positive moments

What steps does a typical person find enjoyable, productive, fun, motivating, delightful, or exciting?

			Easy ticket creation	Only Caller and Short Description to fill in No need to manually select category/subcategory	Less effort on the form for users	Less manual effort for IT staff			Consistent classification, less human error		Automatic classification	Faster ticket routing	Immediate email confirmation			Consistent ticket information			No-code solution that is easy to maintain

Negative moments

What steps does a typical person find frustrating, confusing, angering, costly, or time-consuming?

		Manual classification before automation	Users may not know the correct category	Incorrect category selection	Multiple requests arrive every day					Time-consuming ticket processing	IT staff review each ticket manually		Possible delays in routing Manual process is not scalable as requests increase	Callers informed about ticket creation manually Time spent identifying the issue type	Manual process is error-prone				

Areas of opportunity

How might we make each step better? What ideas do we have? What have others suggested?

How might we make it easy to describe issues with clear keywords?	How might we add more keyword-based classifications?	How might we add more ticket categories?	How might we improve automation for more request types? Could Category and Subcategory choices grow with new issues?							How might we add more Else If conditions for new keywords?			How might we automatically assign tickets to IT groups? Could the flow also fill Assigned group and Assigned to? How might we route tickets to IT staff faster?	Could SLA tracking show how quickly tickets are handled?	How might we keep the solution easy to maintain as it grows?	How might we add predictive intelligence in future? How might we go beyond keyword rules over time?			

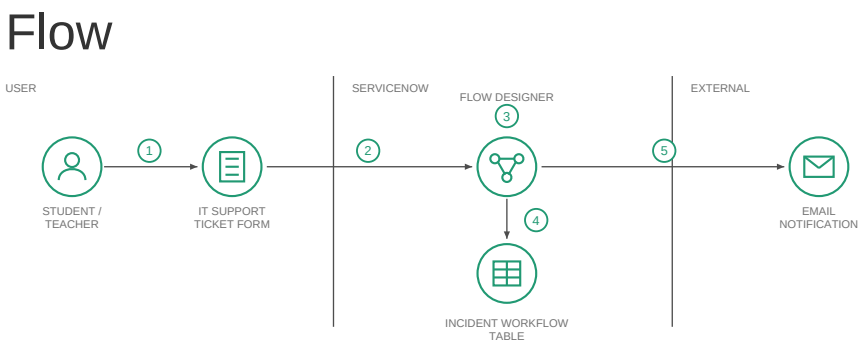
Project Design Phase-II
Data Flow Diagram & User Stories

Date	30 September 2026
Team ID	SWTID-2026-7065
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	4 Marks

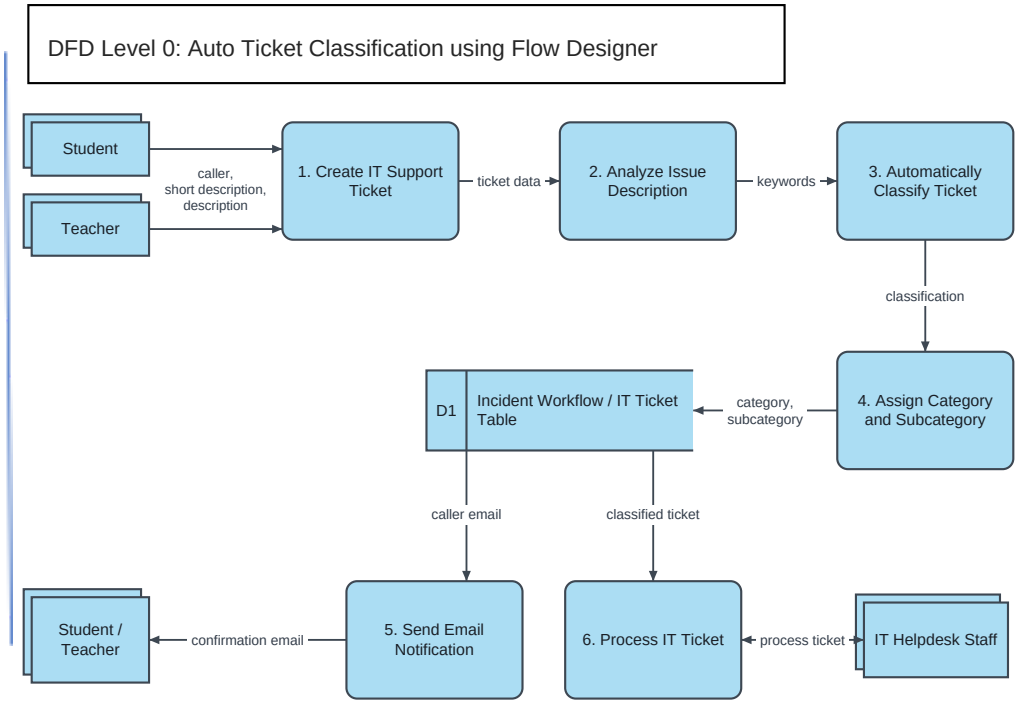
Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: (Simplified)



1. Student or teacher creates an IT support ticket with the caller and a short description.
2. Flow Designer runs when a new Incident Workflow record is created with an empty Category.
3. The flow checks the short description for keywords: WiFi/Network, Projector, Password/Login, Slow/Hanging.
4. Category and Subcategory are set: Network → Wi-Fi, Hardware → Projector, Access → Forgot Password, Performance → Slow Computer.
5. Flow Designer sends the caller an email confirming the request was submitted.



User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Student	Ticket Creation	USN-1	As a student, I can create an IT support ticket by entering my name as the caller.	Ticket is created successfully.	High	Sprint-1
Teacher	Ticket Creation	USN-2	As a teacher, I can enter a short description of my IT problem when I create a ticket.	Short description is captured.	High	Sprint-1
Student	Automatic Classification	USN-3	As a student, I want my ticket to be classified automatically from the short description, so I do not have to choose a category.	Ticket is classified automatically after it is created.	High	Sprint-1
Student	Automatic Classification	USN-4	As a student, I want the Category assigned automatically (Network, Hardware, Access or Performance).	Category is automatically assigned.	High	Sprint-1
Teacher	Automatic Classification	USN-5	As a teacher, I want the Subcategory assigned automatically (Wi-Fi, Projector, Forgot Password or Slow Computer).	Subcategory is automatically assigned.	High	Sprint-1
IT Administrator	Dependent Choices	USN-6	As an IT administrator, I can set dependent Category and Subcategory choices.	Correct Category/Subcategory combination is displayed.	Medium	Sprint-1
Student	Email Notification	USN-7	As a student, I receive an automated email confirming my request was submitted.	Confirmation email is sent to the caller.	High	Sprint-2
Teacher	Automatic Classification	USN-8	As a teacher, I can report Wi-Fi, projector, password or slow computer issues and get them classified.	Correct Category/Subcategory combination is displayed.	High	Sprint-1
IT Administrator	Ticket Storage	USN-9	As an IT administrator, I want ticket details stored in the Incident Workflow table in a structured format.	Ticket information is stored correctly.	High	Sprint-1
IT Helpdesk Staff	Ticket Processing	USN-10	As IT helpdesk staff, I can open a classified ticket and update its State, Assigned group and Assigned to.	IT staff can access and process the ticket.	High	Sprint-2
IT Helpdesk Staff	Ticket Processing	USN-11	As IT helpdesk staff, I can see the Category and Subcategory on each ticket to route it faster.	Category and Subcategory are shown on the ticket.	Medium	Sprint-2
IT Administrator	Maintenance & Scalability	USN-12	As an IT administrator, I can update the classification flow without coding so it is easy to maintain.	Flow can be updated without code.	Low	Sprint-2

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	31 January 2025
Team ID	SWTID-2026-7065
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	IT Ticket Creation	Create a new IT support ticket. Capture Caller information. Capture Short Description. Capture Description. Generate a unique ticket number.
FR-2	Automatic Ticket Classification	Analyze the Short Description. Identify predefined issue keywords. Automatically classify the IT ticket. Eliminate the need for manual classification.
FR-3	Category and Subcategory Assignment	Automatically assign Category. Automatically assign Subcategory. Maintain dependent Category and Subcategory choices. WiFi / Network → Network → Wi-Fi Projector → Hardware → Projector Password / Login → Access → Forgot Password Slow / Hanging → Performance → Slow Computer
FR-4	Email Notification	Send an automated email to the caller after ticket creation. Confirm that the request has been submitted. Provide ticket confirmation information.
FR-5	Structured Ticket Storage	Store ticket information in the Incident Workflow table. Store Caller information. Store Category and Subcategory. Store Short Description and Description. Store State. Store Assigned Group and Assigned To information.
FR-6	Ticket Processing	Allow IT helpdesk staff to access classified tickets. Process tickets based on the assigned category and subcategory. Maintain standardized ticket information.
FR-7	Maintainability and Future Scalability	Keep the solution easy to maintain. Allow additional issue classifications to be added. Support future automation enhancements.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The ticket creation process should be simple and easy for students and teachers to use.
NFR-2	**Security**	Only authorized users should be able to access or modify ticket information.
NFR-3	**Reliability**	The system should consistently classify tickets according to the configured rules and store the information accurately.
NFR-4	**Performance**	Ticket classification and notification should be performed efficiently after ticket creation.
NFR-5	**Availability**	The IT ticketing system should be available for users to submit support requests when required.
NFR-6	**Scalability**	The solution should allow additional categories, subcategories, automation rules, and future features to be added easily.

Project Design Phase-II
Technology Stack (Architecture & Stack)

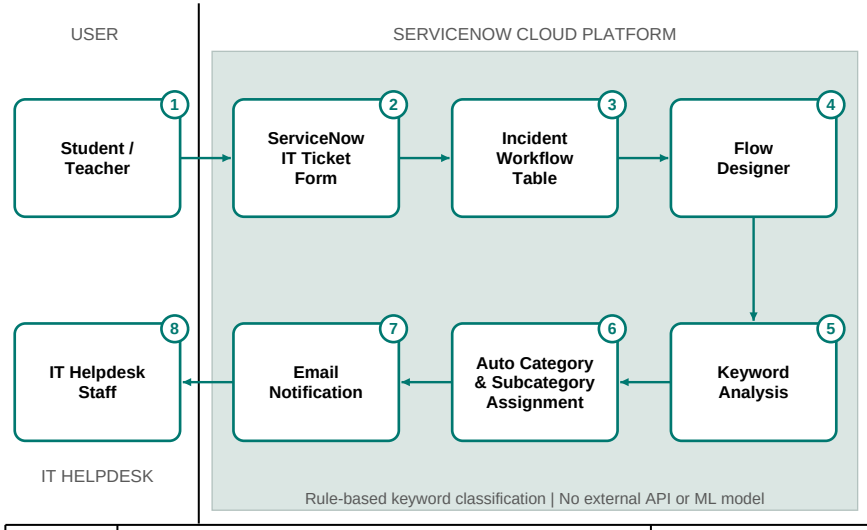
Date	30 September 2026
Team ID	SWTID-2026-7065
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Auto Ticket Classification using Flow Designer (ServiceNow School IT Helpdesk)

Reference: https://docs.servicenow.com



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

S.No	Component	Description	Technology
1.	User Interface	Interface used by students and teachers to create IT support tickets and enter issue details.	ServiceNow Ticket/Form Interface

2.	Application Logic-1	Automatically analyzes the ticket short description and identifies predefined issue keywords.	ServiceNow Flow Designer
3.	Application Logic-2	Automatically assigns the appropriate Category and Subcategory based on the identified issue.	Flow Designer If / Else If Conditions and Update Record Actions
4.	Application Logic-3	Sends an automated email notification to the caller after ticket creation and classification.	ServiceNow Flow Designer Send Email Action
5.	Database	Stores structured IT ticket information including caller, category, subcategory, description, state, assigned group and assigned user.	ServiceNow Incident Workflow / Custom Table
6.	Cloud Database	Not separately used in the project because the solution uses the ServiceNow platform for data storage.	ServiceNow Platform Database
7.	File Storage	Stores project/update-set export files when required for deployment or transfer.	ServiceNow Update Set XML / Local File Storage
8.	External API-1	No external API is required for the ticket classification process.	Not Used
9.	External API-2	No external API is required for the ticket classification process.	Not Used
10.	Machine Learning Model	The project uses predefined keyword-based Flow Designer conditions instead of a machine learning model.	Not Used — Rule-Based Classification
11.	Infrastructure (Server / Cloud)	The application is developed and executed within the ServiceNow platform. Local Server Configuration: Not required Cloud Server Configuration: ServiceNow cloud instance	ServiceNow Cloud Platform

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	The solution is implemented using ServiceNow Flow Designer and does not depend on external open-source frameworks.	ServiceNow Flow Designer
2.	Security Implementations	Access to ticket information and modification of records should be restricted to authorized users and appropriate ServiceNow roles.	ServiceNow Roles and Access Controls
3.	Scalable Architecture	The solution can be extended by adding new categories, subcategories, keywords, automation rules and additional workflow actions without redesigning the entire system.	ServiceNow Flow Designer
4.	Availability	The solution operates on the ServiceNow platform, allowing users to create and manage IT support tickets through the available ServiceNow environment.	ServiceNow Cloud Platform
5.	Performance	Automatic classification occurs when a ticket is created, reducing manual classification work and improving the efficiency of ticket routing.	ServiceNow Flow Designer Automation

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d

Problem-Solution fit canvas 2.0

PTou rapuotsoem /a Vticisailolyn classify school IT helpdesk tickets at creation using ServiceNow Flow Designer (no-code).

Define CS, fit into CC

1. CUSTOMER SEGMENT(S) — CS

Whoisyour customer?
i.e. working parents of 0-5 y.o. kids

- **Students and teachers** of the school who raise IT support tickets (Wi-Fi, projector, password and slow-computer issues).
- **School IT helpdesk agents** who receive, review and route these tickets.

6. CUSTOMER CONSTRAINTS — CC

Whatconstraints prevent yourcustomers from taking action or limit their choices of solutions? i.e. spending power, budget, no cash, network connection, available devices.

- Limited IT staff available to review every incoming request manually.
- Tickets must be classified correctly using standard Category and Subcategory values (dependent choices).
- The solution must be no-code and easy for the school to maintain. Users need confirmation that their request has been submitted. Process must handle growing request volume without extra effort.

5. AVAILABLE SOLUTIONS — AS

Which solutions areavailable tothecustomers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? i.e. pen and paper is an alternative to digital notetaking

- **Manual review:** IT staff read each ticket and select the category and subcategory by hand.

Pros: human judgement. Cons: time-consuming, error-prone, inconsistent and not scalable as requests increase.

- **Informing callers manually** that a ticket has been created – adds extra effort for agents.
- **Our solution:** Flow Designer automation classifies tickets on creation and sends an automatic email.

Explore AS, differentiate

Focus on J&P, tap into BE, understand RC

2. JOBS-TO-BE-DONE / PROBLEMS — J&P

Which jobs-to-be-done(or problems)do youaddress for your customers? There could be more than one; explore different sides.

- Classify each incident automatically when it is created (Network / Hardware / Access / Performance).
- Reduce manual effort for IT agents.
- Improve ticket routing efficiency and consistency.
- Notify the caller that the request has been submitted.
- Provide a no-code, easily maintainable solution.

9. PROBLEM ROOT CAUSE — RC

What is the real reason that thisproblem exists?
What is the back story behind the need to do this job?
i.e. customers have to do it because of the change in regulations.

- The school IT helpdesk receives many incident requests daily from students and teachers on common issues: network connectivity, hardware failures, account access and system performance.
- Categories are assigned manually by IT staff, so the process is slow, error-prone and does not scale as the number of requests grows.

7. BEHAVIOUR — BE

Whatdoes yourcustomer do to address the problem and get the job done?
i.e. directly related: find the right solar panel installer, calculate usage and benefits; indirectly associated: customers spend free time on volunteering work (i.e. Greenpeace)

- Students/teachers create an IT support ticket in the **Incident Workflow** table and enter a Short Description (e.g. “WiFi not working in library”).
- IT agents currently open each ticket, identify the issue type and manually select the Category and Subcategory.
- Agents then inform the caller that the ticket has been created.

Focus on J&P, tap into BE, understand RC

Identify strong TR & EM

3. TRIGGERS — TR

What triggers customers to act? i.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news.

- A student or teacher creates a new ticket (Record Created) with the Category field left empty – this triggers the flow.
- Frequent Wi-Fi, projector, password and slow-computer issues reported by students and teachers.

4. EMOTIONS: BEFORE / AFTER — EM

Howdo customers feelwhen theyface a problem or a job and afterwards?
i.e. lost, insecure > confident, in control - use it in your communication strategy & design.

- **Before:** unsure whether the request was received; agents spend time on manual review.
- **After:** reassured by the confirmation email; tickets are classified correctly and agents can focus on resolving.

10. YOUR SOLUTION — SL

If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality.
If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour.

Flow: “Auto Classify School IT Tickets” (ServiceNow Flow Designer)

- **Trigger:** Record Created on Incident Workflow; condition: Category is Empty.
- If / Else If conditions check keywords in the Short Description:
- WiFi/Network → Network / Wi-Fi; Projector → Hardware / Projector; Password/Login → Access / Forgot Password; Slow/Hanging → Performance / Slow Computer.
- Update Record sets Category and Subcategory; an automated email is sent to the caller.
- Maintained via Update Set (In progress → Complete, exportable as XML).

8. CHANNELS of BEHAVIOUR — CH

8.1 ONLINE

What kind of actions do customers take online? Extract online channels from #7

- Ticket created online in ServiceNow (Incident Workflow form).
- Flow runs automatically; confirmation email sent to caller.

8.2 OFFLINE

What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development.

- Offline (current process): IT agents manually review each request and assign the category and subcategory.
- Agents inform callers about ticket creation manually.

Extract online & offline CH of BE

Problem-Solution it canvas is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 license
Created by Daria Nepriakhina / Amaltama.com

AMALTAMA

Project Design Phase
Problem – Solution Fit Template

Date	30 September 2026
Team ID	SWTID-2026-7065
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	2Marks

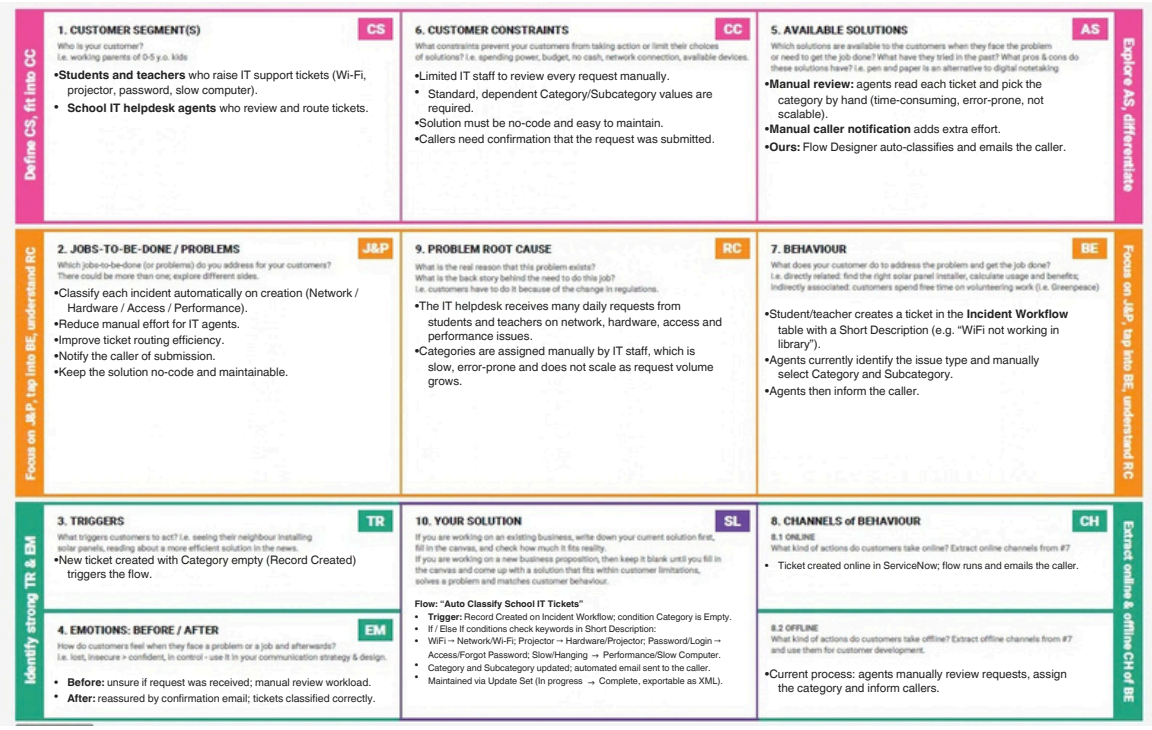
Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ **Understand the existing situation in order to improve it for your target group.**

Template:



References:

1. https://www.ideahackers.network/problem-solution-fit-canvas/
2. https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe

Project Design Phase
Proposed Solution Template

Date	30 September 2026
Team ID	SWTID-2026-7065
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	The school IT helpdesk receives many requests from students and teachers (Wi-Fi, projector, password and slow computer issues). IT staff review each ticket manually to assign a category, which is time-consuming and inefficient.
2.	Idea / Solution description	A ServiceNow Flow Designer flow, Auto Classify School IT Tickets, runs when an Incident Workflow record is created with an empty Category. It checks keywords in the Short Description, sets Category and Subcategory, and emails the caller.
3.	Novelty / Uniqueness	A no-code, easily maintained solution using If / Else If keyword rules and dependent Category and Subcategory fields, with an automatic confirmation email. No scripting or machine learning is used.
4.	Social Impact / Customer Satisfaction	Tickets are classified consistently at creation, reducing manual effort for IT agents and improving routing. Callers receive an automatic email confirming their request has been submitted.
5.	Business Model (Revenue Model)	Not applicable - internal school IT helpdesk automation built on ServiceNow. No revenue model is defined in the project document.
6.	Scalability of the Solution	Easy to maintain, since keyword rules can be added in Flow Designer. Future scope: assignment automation, SLA tracking and predictive intelligence.

Project Design Phase
Solution Architecture

Date	30 September 2026
Team ID	SWTID-2026-7065
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram:

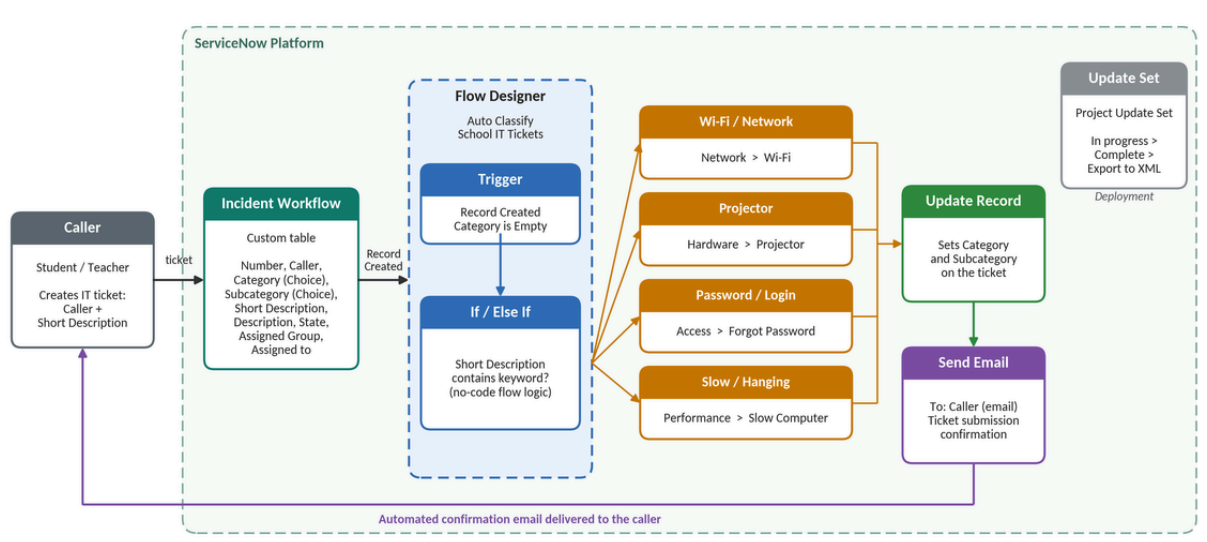


Figure 1: Architecture and data flow of the Auto Ticket Classification solution

Reference:Project document – Auto Ticket Classification using Flow Designer (ServiceNow)

Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story Points)

Date: 30 September 2026 **Team ID:** SWTID-2026-7065 **Maximum Marks:** 5 Marks

Project Name: Auto Ticket Classification using Flow Designer **Team Size:** 5 **Team Leader:** Kirubakaran S

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule.

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Requirement Analysis & Planning	USN-1	As a project team, we can document the business use case and requirements for auto ticket classification.	2	High	Kirubakaran S
Sprint-1	Update Set Creation	USN-2	As a developer, I can create the Project Update Set and make it current.	1	High	Dinesh B
Sprint-1	Custom Table Creation	USN-3	As a developer, I can create the Incident WorkFlow table with Auto Number.	2	High	Sadasivakrishnan R S
Sprint-1	Field Creation and Data Type Configuration	USN-4	As a developer, I can create the ticket fields and choice lists with correct data types.	2	High	Sadasivakrishnan R S
Sprint-2	Category and Subcategory Dependency	USN-5	As a user, I can see Subcategory options based on the selected Category.	2	High	Vijay Ragavan S
Sprint-2	Flow Designer Auto Classification	USN-6	As an IT agent, I get tickets classified automatically by a flow triggered when a ticket is created.	3	High	Dinesh B
Sprint-2	Wi-Fi Issue Classification	USN-7	As a student, my Wi-Fi/Network ticket is set to Network > Wi-Fi.	2	High	Kirubakaran S
Sprint-2	Projector Issue Classification	USN-8	As a teacher, my Projector ticket is set to Hardware > Projector.	2	High	Dinesh B
Sprint-3	Password Issue Classification	USN-9	As a user, my Forgot Password ticket is set to Access > Forgot Password.	2	Medium	Sadasivakrishnan R S
Sprint-3	Slow Computer Classification	USN-10	As a user, my Slow Computer ticket is set to Performance > Slow Computer.	2	Medium	Vijay Ragavan S
Sprint-3	Email Notification	USN-11	As a caller, I will receive a confirmation email once my ticket is created.	2	High	Kirubakaran S
Sprint-4	Testing and Validation	USN-12	As a tester, I can validate the data and run the Wi-Fi and Projector test scenarios.	2	High	Ragulraj H
Sprint-4	Deployment and Final Documentation	USN-13	As a developer, I can complete and export the update set and finalize the documentation.	2	High	Kirubakaran S

Source: Original uploaded Project Planning Template.

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	02 Feb 2026	07 Feb 2026	20	07 Feb 2026
Sprint-2	20	6 Days	09 Feb 2026	14 Feb 2026	20	14 Feb 2026
Sprint-3	20	6 Days	16 Feb 2026	21 Feb 2026	20	21 Feb 2026
Sprint-4	20	6 Days	23 Feb 2026	28 Feb 2026	20	28 Feb 2026

Velocity:
Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{sprint\ duration}{velocity} = \frac{20}{10} = 2$$

Burndown Chart:

A burndown chart is a graphical representation of the work remaining versus time. It is used to track the progress of the Auto Ticket Classification using Flow Designer project across different sprints.

For this project, the burndown chart represents the completion of tasks related to:

- Requirement Analysis and Planning
- ServiceNow Backend Configuration
- Category and Subcategory Dependency
- Flow Designer Automation
- Email Notification
- Testing and Validation
- Deployment and Documentation

The chart shows how the remaining Story Points decrease as the team completes the planned tasks during each sprint.

References:

https://www.atlassian.com/agile/project-management

https://www.atlassian.com/agile/tutorials/sprints

https://www.atlassian.com/agile/tutorials/epics

https://www.atlassian.com/agile/tutorials/burndown-charts

https://www.atlassian.com/agile/project-management/estimation

A Sprint fixed period or duration in which a team works to complete a set of tasks

An **Epic** is a **big task or project** that is too large to complete in one sprint. It is broken down into **smaller tasks (stories)** that can be completed over multiple sprints.

A **Story** is a small task . It is part of an **Epic**.

A **Story Point** is a number that represents how much effort a story takes to complete. (usually in form of Fibonacci series)

1- Very Easy task
2- Normal task
3- Moderate task
5- Difficult task

Storypoint -5 (1,2,3,5)

Sprint 1

Requirement Analysis & Planning (Epic 1)

Analyzing Manual Ticket Classification Process(USN1) 3

Identifying Business Requirements(USN2) 2

Defining Ticket Classification Rules(USN3) 3

Defining Notification and Data Storage Requirements(USN4) 2

ServiceNow Backend Configuration (Epic 2)

Creating Incident Workflow Table(USN5) 3

Configuring Ticket Fields(USN6) 3

Configuring Field Data Types(USN7) 2

Configuring Choice Values(USN8) 2

Total Story Point in Sprint 1= 3+2+3+2+3+3+2+2=20

Sprint 2

Category and Subcategory Dependency (Epic 3)

Configuring Category as Dependent Field(USN9) 3

Configuring Subcategory Dependency(USN10) 3

Mapping Wi-Fi to Network(USN11) 1

Mapping Projector to Hardware(USN12) 1

Mapping Forgot Password to Access(USN13) 1

Mapping Slow Computer to Performance(USN14) 1

Flow Designer Automation (Epic 4)

Creating Auto Classify School IT Tickets Flow(USN15) 2

Configuring Record Created Trigger(USN16) 2

Checking Whether Category is Empty(USN17) 1

Detecting Wi-Fi/Network Keywords(USN18) 3

Detecting Projector/Hardware Keywords(USN19) 2

Total Story Point in Sprint 2= 3+3+1+1+1+1+2+2+1+3+2=20

Sprint 3

Flow Designer Automation (Epic 4 - Continued)

Detecting Password/Login Keywords(USN20) 3

Detecting Slow/Hanging Keywords(USN21) 3

Automatically Updating Category and Subcategory(USN22) 5

Email Notification (Epic 5)

Configuring Send Email Action(USN23) 3

Sending Confirmation Email to Caller(USN24) 3

Configuring Email Subject and Ticket Confirmation Message(USN25) 3

Total Story Point in Sprint 3= 3+3+5+3+3+3=20

Sprint 4

Testing and Validation (Epic 6)

Testing Wi-Fi Ticket Classification(USN26) 2

Testing Projector Ticket Classification(USN27) 2

Testing Password/Login Ticket Classification(USN28) 2

Testing Slow Computer Ticket Classification(USN29) 2

Verifying Category and Subcategory Values(USN30) 2

Verifying Email Notification(USN31) 2

Deployment and Documentation (Epic 7)

Completing the Update Set(USN32) 1

Exporting the Update Set(USN33) 1

Preparing Final Project Documentation(USN34) 3

Validating Complete End-to-End Workflow(USN35) 3

Total Story Point in Sprint 4= 2+2+2+2+2+2+1+1+3+3=20

Total Story Points

Sprint 1 = 20

Sprint 2 = 20

Sprint 3 = 20

Sprint 4 = 20

Velocity= Total Story Points Completed/ Number of Sprints

Total story Points= 20+20+20+20=80

No of Sprints= 4

Velocity =80/4

20(Story Points per Sprint)

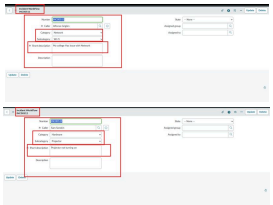
Your team's velocity is 20 Story Points per Sprint.

Project Development Phase
Model Performance Test

Date	30 September 2026
Team ID	SWTID-2026-7065
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	10 Marks

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot
1.	Metrics	Regression Model: Not applicable (rule-based solution, no ML model). Classification Model: Rule-based keyword classification using Flow Designer (If / Else If on Short Description). Confusion Matrix, Accuracy Score and Classification Report: Not applicable. Verified with 2 test scenarios (Wi-Fi, Projector) - both classified as expected.	
2.	Tune the Model	Hyperparameter Tuning - Not applicable (no ML model). Validation Method - Functional testing with sample tickets; confirmation email checked in System Logs > Emails.	

Functional & Performance Testing Template

Model Performance Test

Date	30 September 2026
Team ID	SWTID-2026-7065
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	

Test Scenarios & Results

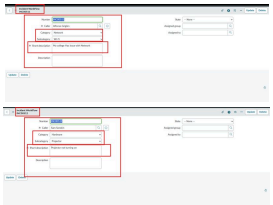
Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
FT-01	Short Description keyword - Wi-Fi / Network (e.g., "WiFi not working in library")	Enter Caller and Short Description; save and reload the form	Category = Network, Subcategory = Wi-Fi	Network / Wi-Fi set	Pass
FT-02	Short Description keyword - Projector (e.g., "Projector not turning on")	Enter Caller and Short Description; save and reload the form	Category = Hardware, Subcategory = Projector	Hardware / Projector set	Pass
FT-03	Email notification to the caller	Search the subject in System Logs > Emails and use Preview Mail	Confirmation email is sent to the caller	Email sent	Pass
FT-04	Category and Subcategory dependency	Open the Incident Workflow form and select a Category	Subcategory choices depend on the selected Category	Choices filtered	Pass
PT-01	Response Time Test	Use a timer to check content generation time	Should be under 3 seconds	Not tested	N/A
PT-02	API Speed Test	Send multiple API calls at the same time	API should not slow down	Not tested	N/A
PT-03	File Upload Load Test (e.g., PDFs)	Upload multiple PDFs and check processing	Should work smoothly without crashing	Not tested	N/A

Project Development Phase
Model Performance Test

Date	30 September 2026
Team ID	SWTID-2026-7065
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	10 Marks

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

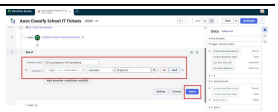
S.No.	Parameter	Values	Screenshot
1.	Metrics	Regression Model: Not applicable (rule-based solution, no ML model). Classification Model: Rule-based keyword classification using Flow Designer (If / Else If on Short Description). Confusion Matrix, Accuracy Score and Classification Report: Not applicable. Verified with 2 test scenarios (Wi-Fi, Projector) - both classified as expected.	
2.	Tune the Model	Hyperparameter Tuning - Not applicable (no ML model). Validation Method - Functional testing with sample tickets; confirmation email checked in System Logs > Emails.	

Project Development Phase
Model Performance Test

Date	30 September 2026
Team ID	SWTID-2026-7065
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

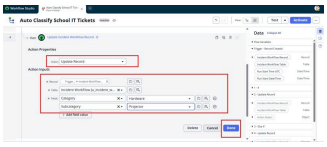
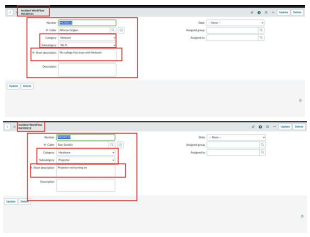
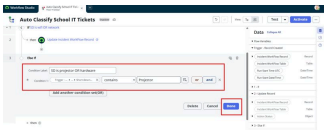
S.No.	Parameter	Screenshot / Values
1.	Data Rendered	Ticket records are stored in the custom Incident Workflow table (Number, Caller, Category, Subcategory, Short Description, Description, State, Assigned Group, Assigned to).
2.	Data Preprocessing	No data preprocessing required. The flow reads the Short Description of each new ticket and matches predefined keywords.
3.	Utilization of Data Filters	Trigger condition: Category is Empty. If / Else If conditions: Short Description contains WiFi / Network, Projector, Password / Login or Slow / Hanging. 
4.	DAX Queries Used	Not applicable - no-code Flow Designer solution; no DAX queries are used.
5.	Dashboard design	No of Visualizations / Graphs - Not applicable (no dashboard in this project)
6	Report Design	No of Visualizations / Graphs - Not applicable (no report in this project)

Project Development Phase
Model Performance Test

Date	30 September 2026
Team ID	SWTID-2026-7065
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

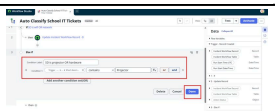
S.No.	Parameter	Values	Screenshot
1.	Model Summary	ServiceNow Flow Designer automation (Auto Classify School IT Tickets) to classify school IT tickets using the custom Incident Workflow table with Choice and Reference fields. Note: When a ticket is created with an empty Category, the flow checks the Short Description for keywords, updates Category and Subcategory, and sends a confirmation email to the caller.	
2.	Accuracy	Training Accuracy - Not applicable (rule-based classification, no ML model) Validation - 2 of 2 test scenarios (Wi-Fi, Projector) classified as expected	
3.	Confidence Score (Only Yolo Projects)	Not applicable - not a YOLO project. Classes assigned by the flow (rule-based, no confidence score): Network / Wi-Fi, Hardware / Projector, Access / Forgot Password, Performance / Slow Computer.	

Project Development Phase
Performance Test

Date	30 September 2026
Team ID	SWTID-2026-7065
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Screenshot / Values
1.	Data Rendered	Ticket records are stored in the custom Incident Workflow table (Number, Caller, Category, Subcategory, Short Description, Description, State, Assigned Group, Assigned to).
2.	Data Preprocessing	No data preprocessing required. The flow reads the Short Description of each new ticket and matches predefined keywords.
3.	Utilization of Filters	Trigger condition: Category is Empty. If / Else If conditions: Short Description contains WiFi / Network, Projector, Password / Login or Slow / Hanging. 
4.	Calculation fields Used	Not applicable - no calculated fields; Category and Subcategory are set by Update Record actions in Flow Designer.
5.	Dashboard design	No of Visualizations / Graphs - Not applicable (no dashboard in this project)
6	Story Design	No of Visualizations / Graphs - Not applicable (no story in this project)

User Acceptance Testing (UAT) Template

Date	30 September 2026
Team ID	SWTID-2026-7065
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	

Project Overview:

Project Name: Auto Ticket Classification using Flow Designer

Project Description: Auto-classification of school IT tickets using ServiceNow Flow Designer

Project Version: [Version Number]

Testing Period: [Start Date] to [End Date]

Testing Scope:

Automatic Category / Subcategory classification from Short Description; email notification to caller

Classify at creation; dependent Category and Subcategory; automated confirmation email

Testing Environment:

URL/Location: ServiceNow instance - Incident Workflow table and Flow Designer

Credentials (if required): [Username/Password]

Test Cases:

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	WiFi keyword ("WiFi not working in library")	1. Enter Caller 2. Enter Short Description 3. Save and reload the form	Network / Wi-Fi set; email sent to caller	Network / Wi-Fi set; email sent to caller	Pass
TC-002	Projector keyword ("Projector not turning on")	1. Enter Caller 2. Enter Short Description 3. Save and reload the form	Hardware / Projector set; email sent to caller	Hardware / Projector set; email sent to caller	Pass

Bug Tracking:

Bug ID	Bug Description	Steps to reproduce	Severity	Status	Additional feedback
BG-001	[Describe the issue or bug encountered	[Step 1] [Step 2] [Step 3]	[Low/Medium/High]	[Open/In Progress/	[Any additional comments or

	]			Closed]	feedback]
...	...	...	...	...	...

Sign-off:

Tester Name: [Name of Tester]

Date: [Date of Test Completion]

Signature: [Tester's Signature]

Notes:

- Ensure that all test cases cover both positive and negative scenarios.
- Encourage testers to provide detailed feedback, including any suggestions for improvement.
- Bug tracking should include details such as severity, status, and steps to reproduce.
- Obtain sign-off from both the project manager and product owner before proceeding with deployment.

Acceptance Testing
UAT Execution & Report Submission

Date	30 September 2026
Team ID	SWTID-2026-7065
Project Name	Auto Ticket Classification using Flow Designer
Maximum Marks	4 Marks

1. Purpose of Document

The purpose of this document is to briefly explain the test coverage and open issues of the Auto Ticket Classification using Flow Designer project at the time of the release to User Acceptance Testing (UAT).

2. Defect Analysis

This report shows the number of resolved or closed bugs at each severity level, and how they were resolved

Resolution	Severity 1	Severity 2	Severity 3	Severity 4	Subtotal
By Design	-	-	-	-	-
Duplicate	-	-	-	-	-
External	-	-	-	-	-
Fixed	-	-	-	-	-
Not Reproduced	-	-	-	-	-
Skipped	-	-	-	-	-
Won't Fix	-	-	-	-	-
Totals	-	-	-	-	-

3. Test Case Analysis

This report shows the number of test cases that have passed, failed, and untested

Section	Total Cases	Not Tested	Fail	Pass
Wi-Fi Classification	1	0	0	1
Projector Classification	1	0	0	1
Password Classification	1	1	0	0
Slow Computer Classification	1	1	0	0

Email Notification	2	0	0	2
Category-Subcategory Dependency	1	0	0	1
-	-	-	-	-